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REVIEW ARTICLE

Molecular and cellular mechanisms underlying gyrate atrophy: Why is the retina primarily affected?

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Abstract

Gyrate atrophy of the choroid and retina (GACR; OMIM #258870) is a rare early-onset autosomal recessive disorder, caused by bi-allelic pathogenic variants in the gene coding for ornithine aminotransferase (*OAT*) resulting in hyperornithinaemia. Clinically, GACR is characterized by the concentric loss of visual fields due to progressive chorioretinal atrophy. Because OAT is systemically expressed, it is not clear why primarily the retina is damaged in GACR patients. In this review, we first provide an extensive overview of the clinical features and current treatment modalities for GACR. Next, we discuss the different pathways involved in ornithine metabolism, including the urea cycle, polyamine synthesis, creatine synthesis, proline synthesis and degradation and provide our vision on how OAT deficiency is thought to affect these pathways in the retinal pigment epithelium (RPE). We provide several hypotheses to explain the retinal pathology observed in GACR and discuss perspectives on future research.

KEYWORDS
gyrate atrophy, inherited metabolic disorder, OAT, ornithine, retinal degeneration

Mark J. N. Buijs and Berith M. Balfoort shared first authorship.

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Prof. Arthur Bergen 1961–2024

This manuscript is dedicated to the legacy of our visionary colleague Prof. Arthur Bergen, who sadly passed away unexpectedly during the final preparation phase of this manuscript.

1 | CLINICAL FEATURES, GENETICS AND CURRENT TREATMENT MODALITIES FOR GACR

1.1 | Clinical features of patients with GACR

Gyrate atrophy of the choroid and retina (GACR) is characterised by progressive constriction of visual fields due to progressive chorioretinal atrophy. The theoretical global incidence of the disorder has been estimated to be around 1:1 500 000 (Montioli, Bellezza, et al., 2021). In Finland, the disease is much more common due to a founder effect and is estimated to affect 1:50000 (Takki, 1974). Although ornithine aminotransferase (OAT) is expressed in, virtually every cell of the body, GACR and thus OAT deficiency is characterized by a predominantly ophthalmic phenotype. Patients typically present within the early decades of life with symptoms such as myopia and nyctalopia. Over time, they experience concentric visual field loss due to progressive chorioretinal atrophy and eventually develop severe visual impairment and functional blindness (Elnahry & Elnahry, 2022; Valle et al., 1980). Upon fundoscopy, characteristic round and sharply demarcated lesions can be seen, which in time coalesce into large atrophic zones (see Figure 1). Central vision can remain intact for several decades, although it can be severely affected by several complications as early (bilateral) cataract, cystoid maculopathy, macular neovascularization and epiretinal membrane formation (Elnahry & Elnahry, 2022; Mansour et al., 2021; Marano et al., 1996; Valle & Simell, 2019). On electroretinography (ERG), scotopic and photopic responses are often impaired and are likely to be entirely extinguished, although exceptions have been reported (Balfoort et al., 2021; Jasani et al., 2018).

Our recent cross-sectional cohort study showed a high prevalence of high myopia, with a spherical equivalent of the refractive error of −8.96, which is considerably higher than what has been reported in other inherited retinal diseases (Balfoort, Van Den Broeck, et al., 2024). High myopia, defined as a refractive error with a spherical equivalent of more than −6.0 dioptres, is associated with retinal and choroidal changes, including decreased vessel density and choroidal blood flow. This likely develops due to axial elongation (Tedja et al., 2018; Wang et al., 2021). This is probably also the case in GACR, but this remains to be elucidated. Nonetheless, myopia is consistently reported in GACR cases, even though precise refractive errors are often not mentioned (Balfoort et al., 2021; Casalino et al., 2018; Christopher et al., 1999; Kaiser-Kupfer et al., 1991; McInnes et al., 1980; Michel et al., 2015; Palmer et al., 2023; Shrestha et al., 2010). Another common complication of inherited retinal diseases as GACR is cystoid maculopathy, or cystoid macular oedema (CME) (Marques et al., 2022). Cystoid maculopathy in GACR may result from a reduced integrity of the outer blood–retina barrier (oBRB) leading to extracellular, intraretinal oedema at the macula. This can lead to further deterioration of visual acuity, but might be managed with therapeutic interventions as oral acetazolamide (Mansour et al., 2021). There is a large degree of phenotypic heterogeneity among GACR patients, both within and between families, independent of the pathogenic variants involved. Genotype–phenotype studies are difficult due to confounders introduced by therapies (i.e. medical diet and supplements) and/or variability in compliance (Brody et al., 1992; Peltola et al., 2001; Sen et al., 2021).

Extra-ophthalmic manifestations of GACR include variable and often subclinical neurological and/or muscular symptoms. Possibly, these neurological symptoms are caused by secondary creatine deficiency, which develops due to the inhibitory effect of high (sub) cellular ornithine levels on arginine: glycine amidinotransferase (AGAT), the first enzyme in creatine biosynthesis. Creatine is a high-energy substrate which is of great importance especially for the brain and muscle (Joncquel-Chevalier et al., 2015). Indeed, primary creatine deficiencies, such as AGAT deficiency, and especially creatine transporter deficiency due to mutations in *SLC6A8*, are associated with profound neurological symptoms (Stockler-Ipsiroglu & van Karnebeek, 2014). Secondary creatine deficiency has been consistently reported in patients with GACR, but associated (sub) clinical symptoms have rarely been reported. A recent systematic review reported neurological symptoms in 9 out of 97 patients with GACR, corresponding to the <10% that is reported in other literature (Balfoort et al., 2021; Valle & Simell, 2019). It should be noted, however, that only in 13 of these patients were symptoms reported beyond the ophthalmic phenotype, and likely not all patients were screened for neurological symptoms.



FIGURE 1 Colour fundoscopy photography of the right eye of a patient with GACR, showing progressive chorioretinal atrophy over time. (a) Fundoscopy image of the central retina of the patient at age 25, immediately after the diagnosis of GACR. There are a few atrophic lesions surrounding the optic nerve. Peripheral GACR lesions are described in the patient status. (b) Fundoscopy image of the central retina of the patient at age 35. There is an increase in size of the atrophic lesions surrounding the optic nerve. Peripheral GACR lesions are described in the patient status. (c) Ultra-widefield (UWF) fundoscopy image of the retina of the patient at age 40. There is progressive peripheral chorioretinal atrophy, with a circular area of central retina left. The retinal lesions have coalesced into large atrophic zones.

Involvement of the peripheral nervous system in GACR has also been described, although this has largely been subclinical (Peltola et al., 2002). One cohort study described 37 GACR patients, of whom three had a history of seizures. Central nervous system abnormalities were described in these patients using magnetic resonance imaging (MRI) and electroencephalogram (EEG) but did not seem to be correlated with clinical symptoms (Valtonen et al., 1999). In another cohort, 5 out of 7 patients suffered from behavioural disorders and/or neurodevelopmental delay, but unfortunately, pathogenic variants were not specified within this cohort, precluding proper interpretation (Valayannopoulos et al., 2009). Why creatine deficiency does not seem to lead to a more profound neurological phenotype in GACR is unclear.

A possible first presentation of GACR is neonatal hyperammonaemia. During the neonatal phase, ornithine is an essential amino acid due to the low initial expression of arginase 1 (encoded by *ARG1*), which explains why OAT flux favours ornithine production instead of ornithine consumption. As ornithine is required for a functional urea cycle, low levels of ornithine may well lead to a secondary urea cycle defect. Treatment with intravenous arginine is usually sufficient to restore the urea cycle (Cleary et al., 2005; de Sain-van der Velden et al., 2012) and without treatment, hyperammonaemia may remain clinically unnoticed and/or resolve spontaneously. Natural history studies are ongoing to further delineate the phenotype and disease course.

1.2 | Genetics of GACR

The genetics of GACR has been reviewed recently (Molaei Ramshe et al., 2024; Montioli, Bellezza, et al., 2021). In summary, the human *OAT* gene is located on chromosome 10 and encodes a 439 amino acid protein (Barrett et al., 1987). Different (shorter) *OAT* transcripts have been described; however, their exact function and tissue distribution are unclear. The existence of an *OAT* splice variant that does not excise exon 2, unlike the canonical variant, has been described only in a retinoblastoma cell line (Fagan et al., 1989). The ClinVar database currently reports 44 likely pathogenic (class 4) and 82 pathogenic (class 5) *OAT* variants (Richards et al., 2015). These variants are located throughout the entire gene and all result in GACR (Floriani et al., 2023). Even though there is no clear genotype–phenotype correlation in GACR, some pathogenic variants are associated with pyridoxine responsiveness (Montioli, Bellezza, et al., 2021). Current bioinformatics and biochemical work regarding the effect of specific pathogenic variants on the function of the OAT enzyme may help to further understand the exact pathogenic mechanism in GACR (Floriani et al., 2023).

1.3 | Diagnosing GACR

In the past, GACR often had been diagnosed based on typical retinal lesions ascertained by an ophthalmologist. Subsequent plasma amino acid measurements showing hyperornithinaemia are required for confirmation of the diagnosis. Currently, *OAT* is increasingly included in genetic screening panels for inherited retinal diseases, facilitating the diagnosis of GACR (Valle & Simell, 2019). In case of variants of unknown significance (VUS), the presence of hyperornithinaemia and a history of neonatal hyperammonaemia and/or retinal lesions can be used to support the pathogenicity of the variants.

Children with GACR who initially present with neonatal hyperammonaemia are often diagnosed with GACR before the onset of ophthalmic symptoms (Cleary et al., 2005; de Sain-van der Velden et al., 2012). Measurements of residual enzymatic OAT activity in fibroblasts may assist in predicting outcomes and can be used to assess pyridoxine responsiveness, which then can be used for therapy.

1.4 | Currently applied treatment modalities for patients with GACR

There are currently no international guidelines or treatment recommendations for individuals with GACR. As a result, management varies between physicians.

This heterogeneity impedes reliable evaluation of the effectiveness of current and future therapeutic modalities. In order to harmonise therapeutic approaches in The Netherlands, our group wrote a multidisciplinary care pathway based on a systematic review of all described treatment modalities up until 2020 (Balfoort et al., 2021).

Overall, therapeutic modalities currently applied in clinical practice are centred around three different targets, which include (1) reduction of plasma ornithine, adhering to the hypothesis that ornithine is toxic to cells, especially those of the choroid and retina, (2) correction of secondary creatine deficiency and (3) supportive management of ophthalmic complications. Evidence for the efficacy of these interventions is scarce, and treatment is often initiated based on biochemical reasoning and small cohort studies, yielding a level of evidence commonly between 4 and 5 according to the Levels of Evidence for Therapeutic Studies of the Center for Evidence-Based Medicine (Burns et al., 2011).

1.4.1 | Therapeutic interventions to lower plasma ornithine in patients

A protein-restricted or arginine-restricted diet is advised for most patients. Through dietary reduction of arginine, which is an ornithine precursor, plasma ornithine may be reduced. Other essential amino acids are supplied through dietary supplement mixtures. Long-term reduction of ornithine seems to slow down retinal degeneration (Kaiser-Kupfer et al., 1991, 2002, 2004). However, despite strict adherence to dietary therapy, normalisation of plasma ornithine is often not accomplished (Balfoort et al., 2021).

OAT requires pyridoxal-5′-phosphate (PLP) as a co-factor to exert its function. Supplementing high doses of the PLP precursor pyridoxine can benefit at least a subgroup of GACR patients by lowering plasma ornithine levels (Cui et al., 2018; Hayasaka et al., 1985; Kennaway et al., 1980; Mashima et al., 1999; Michaud et al., 1995; Ohkubo et al., 2005). These patients are considered pyridoxine-responsive. The percentage of pyridoxine-responsive GACR patients is estimated at around 5%, although a recent systematic review reported pyridoxine responsiveness in 30% of patients (Balfoort et al., 2021; Valle & Simell, 2019). Pyridoxine responsiveness is often determined based on plasma ornithine levels after oral administration of pyridoxine. We have developed an in vitro assay to test pyridoxine responsiveness in primary fibroblasts of GACR patients (Balfoort, Pampalone, et al., 2024).

L-lysine supplementation has been attempted in GACR patients to reduce plasma ornithine levels. Because ornithine, lysine and arginine share a common renal transport mechanism, increasing plasma lysine levels is predicted to competitively inhibit ornithine reabsorption in the kidney. Two studies attempted to supply GACR patients with oral L-lysine and reported a reduction of 21%–31% of plasma ornithine upon supplementation of 10–15 g/day (Elpeleg & Korman, 2001; Peltola et al., 2000). To our knowledge, L-lysine supplementation has not been described in combination with other therapeutic modalities. Pyridoxine, creatine and a natural protein-restricted diet have been used concurrently.

1.4.2 | Therapeutic interventions to correct secondary creatine deficiency

Secondary creatine deficiency is commonly treated with supplementation of creatine or creatine precursors (methionine and guanidinoacetic acid (GAA)), which was found to partially restore creatine peaks in brain and muscle as assessed by magnetic resonance spectroscopy in GACR patients. However, these results have not been correlated with clinical outcome measures (Heinanen et al., 1999; Nanto-Salonen et al., 1999). Creatine is frequently supplemented to patients with GACR, but treatment initiation often coincides with the start of other treatment modalities. Moreover, in most cases no clinical (neurological) outcome measures have been specified before initiation of therapy (Balfoort et al., 2021). Together, this makes it difficult to conclude whether or not creatine supplementation is favourable and what the effect is on biochemical and/or clinical outcome parameters.

1.4.3 | Treatment of ophthalmic complications related to GACR

Ophthalmic complications of GACR include cataract, cystoid maculopathy and high myopia. Less commonly reported abnormalities are pigment abnormalities and neovasularisation. Each of these ophthalmic complications may require therapeutic intervention.

Cataract is prevalent in patients with inherited (metabolic) retinal diseases, and patients tend to require surgery at a younger age than patients with age-related cataract (Khojasteh et al., 2023; Nguyen et al., 2023). Patients with GACR require cataract surgery, often already in their late twenties (Balfoort, Van Den Broeck, et al., 2024; Palmer et al., 2023). The reason for the development of early cataract in these patients is unclear.

GACR patients frequently have cystoid maculopathy, or CME, which is commonly treated with carbonic anhydrase inhibitors (Catala-Mora et al., 2024). The efficacy of treatment with carbonic anhydrase inhibitors in patients with GACR is variable, and the outcomes of therapy are not always mentioned. One study reported an improvement in anatomical outcome measures in patients treated for cystoid maculopathy with carbonic anhydrase inhibitors. However, many treated patients retained residual cystoid maculopathy (Balfoort, Van Den Broeck, et al., 2024).

A recent cohort study reported on therapy with oral carbonic anhydrase inhibitors in one third of their GACR patient cohort, but whether this had any effect was not reported (Palmer et al., 2023). Several case reports mention possible efficacy of topical and/or oral administration of carbonic anhydrase inhibitors (Cavdarli et al., 2020; Piozzi et al., 2017; Zekusic et al., 2018). The effect of carbonic anhydrase inhibitors

on cystoid maculopathy has been attributed to decreased leakage and stimulated active transport across the oBRB in retinitis pigmentosa patients (Moldow et al., 1998). The retinal pigment epithelium (RPE) is an important aspect of the oBRB and is involved in GACR pathogenesis (Kaneko, Ando, et al., 2007; Nakauchi et al., 2003; Wilson et al., 1991). This might explain the varying efficacy of carbonic anhydrase inhibitors in the treatment of cystoid maculopathy in GACR.

Other therapeutic interventions that have been attempted in order to resolve cystoid maculopathy in patients with GACR are topical non-steroidal anti-inflammatory drops (Cavdarli et al., 2020; Piozzi et al., 2017), intravitreal or periocular corticosteroids (Abdelmassih et al., 2019; Alparslan et al., 2018; Vasconcelos-Santos et al., 2007), or intravitreal anti-vascular endothelial growth factor agents (Elnahry et al., 2018; Inanc et al., 2018). Octreotide, a somatostatin analogue, has been used in two patients with GACR and was stopped, likely due to inadequate efficacy or possible side effects (Balfoort, Van Den Broeck, et al., 2024). A positive effect of octreotide has been reported in patients with post-surgical CME (Shah et al., 2010), CME in uveitis (Missotten et al., 2007) and diabetic macular oedema (Hernaez-Ortega et al., 2004).

1.4.4 | Other therapeutic modalities administered in patients with GACR

Proline is a non-essential amino acid and an important structural component of collagen. It is produced from δ1-pyrroline-5-carboxylate (P5C), which is either produced from glutamate via the enzyme pyrroline-5-carboxylate synthetase (P5CS) or from ornithine via OAT (see Figure 5) (Vettore et al., 2021). Only one case series describes oral supplementation of proline in GACR patients. Visual acuity improved in one patient upon proline supplementation, but this was not further specified (Hayasaka et al., 1985). The potential effect of proline on RPE health is described in more detail in Section 3.3.2 below.

1.4.5 | Visual rehabilitation for patients with GACR

In addition to interventions aimed at delaying vision loss, there is also an important role for visual rehabilitation in clinical practice. Low-vision rehabilitation services (LVRSs) implement a multidisciplinary approach to help patients manage the consequences of their disease and lead a lifestyle that is as autonomous as possible. These services can be approached at different times during the course of the disease, and their assistance can vary based on patient needs. Several studies have shown a beneficial effect of LVRSs on perceived quality of life in visually impaired adults (Lamoureux et al., 2007; Langelaan et al., 2009; Stelmack et al., 2012; van Nispen et al., 2020). As visual impairment can currently not be prevented in patients with GACR, it is important to recognize the possibility of offering LVRSs, starting at an early stage of the disease.

2 | UNDERSTANDING GACR: INTRA-CELLULAR ORNITHINE METABOLISM

In this section, we will elaborate on the different pathways in which ornithine plays an important role and discuss how they are affected in GACR. These pathways include the urea cycle, polyamine synthesis, creatine synthesis and proline synthesis and degradation.

2.1 | The urea cycle

The urea cycle and its relationship with ornithine metabolism is depicted in Figure 2. The main goal of the urea cycle, which only occurs in the liver, is to convert ammonia to urea. Ammonia is formed from (dietary) protein catabolism (Barmore et al., 2024). The first steps in the urea cycle occur in the mitochondria (Figure 2). In the hepatocytes, ammonia and bicarbonate are converted to carbamoyl phosphate via carbamoyl phosphate synthetase 1 (CPS1). CPS1 requires the activator N-acetyl-l-glutamate (NAG) in order to perform its function. NAG is synthesized from glutamate and acetyl coenzyme-A by N-acetylglutamate synthase (NAGS). Carbamoyl phosphate combines with ornithine to form citrulline in a reaction that is catalysed by ornithine transcarbamylase (OTC). Citrulline is transported to the cytosol via SLC25A15, where it reacts with aspartate to form argininosuccinate. This reaction is catalysed by the enzyme argininosuccinate synthetase (ASS). Argininosuccinate lyase (ASL) breaks down argininosuccinate in a process that generates arginine and fumarate. Finally, arginase converts arginine into urea and ornithine. There are two arginase isoforms, of which arginase 1 (ARG1) is localized in the cytosol and predominantly expressed in the liver, whereas arginase 2 (ARG2) is localized in mitochondria (Caldwell et al., 2018). The ornithine that is formed from arginine can be transported back into the mitochondria via ORNT1 (*SLC25A15*) or ORNT2 (*SLC25A2*) in exchange for citrulline. Arginine can also be used as a substrate for nitric oxide synthase (NOS), generating nitric oxide and citrulline (Niu et al., 2022).

Genetic defects affecting each of the steps involved in the urea cycle have been reported in the literature including ornithine transcarbamylase deficiency, carbamylphosphate synthetase 1 deficiency and hyperornithinaemia, hyperammonaemia, homocitrullinuria syndrome (HHH syndrome). These deficiencies typically result in hyperammonaemia, and patients commonly suffer from neurological complaints and lethargy. However, unlike GACR, vision-related problems are not typically associated with this group of disorders (Summar & Mew, 2018). The reason for this phenotypic discrepancy among inherited disorders of ornithine metabolism still needs to be elucidated.

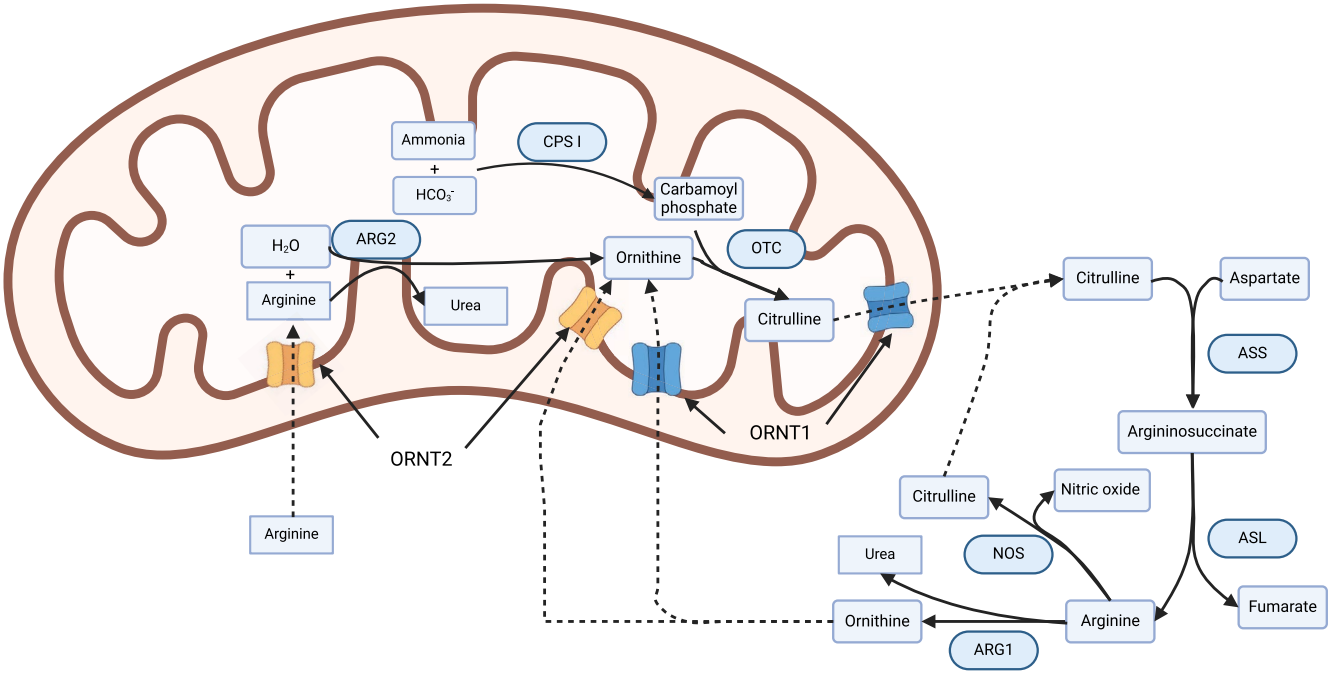


FIGURE 2 Overview of the urea cycle. The urea cycle starts in the mitochondria where ammonia and bicarbonate are converted to carbamoyl phosphate by CPS I (top-middle). Following this, OTC catalyses the formation of citrulline from carbamoyl phosphate and ornithine. Citrulline is transported to the cytosol where it reacts with aspartate to form argininosuccinate in a reaction catalysed by ASS (bottom-right). Argininosuccinate is subsequently converted to arginine and fumarate by ASL. Arginine can either be used to form ornithine and urea via ARG1 or generate nitric oxide and citrulline via NOS. Ornithine can be transported to the mitochondria to serve as a substrate for subsequent urea cycles. Arginine can also be converted to ornithine within the mitochondria via ARG2 in a reaction that also generates urea. Arrows indicate enzymatic reactions, dashed lines indicate transport of compounds. ARG1, arginase 1; ARG2, arginase 2; ASL, argininosuccinate lyase; ASS, argininosuccinate synthetase; CPS1, carbamoyl phosphate synthetase I; HCO_3^-, bicarbonate; NOS, nitric oxide synthase; OTC, ornithine transcarbamylase. Created in https://BioRender.com.

Taken together, there is little evidence to support the involvement of the urea cycle in the retinal abnormalities observed in GACR. Even though ornithine plays an essential role in the urea cycle, the rate-limiting step of this process is the conversion of ammonia and bicarbonate to carbamoyl phosphate by CPS1 (Barmore et al., 2024), and hence hyperornithinaemia as a result of GACR will not lead to increased urea cycle activity. Furthermore, the urea cycle occurs in the liver, which is unaffected in GACR patients, and not within the retina, again suggesting that the urea cycle is likely not directly involved in the pathogenesis of GACR. Moreover, OAT itself is expressed in perivenous hepatocytes and not in periportal hepatocytes, which are the cell types that express the urea cycle enzymes (Ginguay et al., 2017). Therefore, OAT dysfunction is unlikely to affect the ornithine pool within periportal hepatocytes. Finally, in a zebrafish model, inhibition of OAT was able to prevent hyperammonaemia-induced biochemical alterations and prevent ammonia-induced lethality (Zielonka et al., 2018). This protective effect of OAT inhibition was mainly attributed to decreased OAT-dependent glutamate production and therefore preventing glutamate excitotoxicity (see also Section 2.4).

2.2 | The polyamine synthesis pathway

An overview of the polyamine synthesis pathway and how this pathway relates to ornithine metabolism is presented in Figure 3. The polyamines putrescine, spermidine and spermine are implicated in many physiological processes, among which are cell proliferation, cell survival, gene regulation, differentiation and generation of oxidative stress. Consequently, dysregulation of the polyamine synthesis pathway has been implicated in a range of diseases. The intracellular level and type of polyamines are tightly regulated (Hirano et al., 2021; Novita Sari et al., 2021).

The first and rate-limiting step of polyamine synthesis is the conversion of cytoplasmic ornithine to putrescine in a reaction that is catalysed by ornithine decarboxylase (ODC) (Figure 3). In addition to the main pathway of polyamine synthesis, an alternative pathway for generating putrescine has been identified, in which arginine is converted to agmatine via arginine decarboxylase (*ADC*), and subsequently converted to putrescine by agmatinase (*AGMAT*) (Wang et al., 2014). Putrescine can then be converted to spermidine by spermidine synthase (*SRM*), from which spermine can be generated via spermine synthase (*SMS*) (Minois et al., 2011) (Figure 3).

Spermine and spermidine can be catabolised in a process that generates reactive oxygen species (ROS). First, spermidine or spermine is acetylated by spermine/spermidine-N-acetyltransferase (*SSAT*) to form N-acetylspermine and N-acetylspermidine, respectively. Once acetylated, polyamine oxidase (*PAOX*) can catalyse the conversion of N-acetylspermine to spermidine and N-acetylspermidine to putrescine. In this oxidation reaction both hydrogen peroxide and

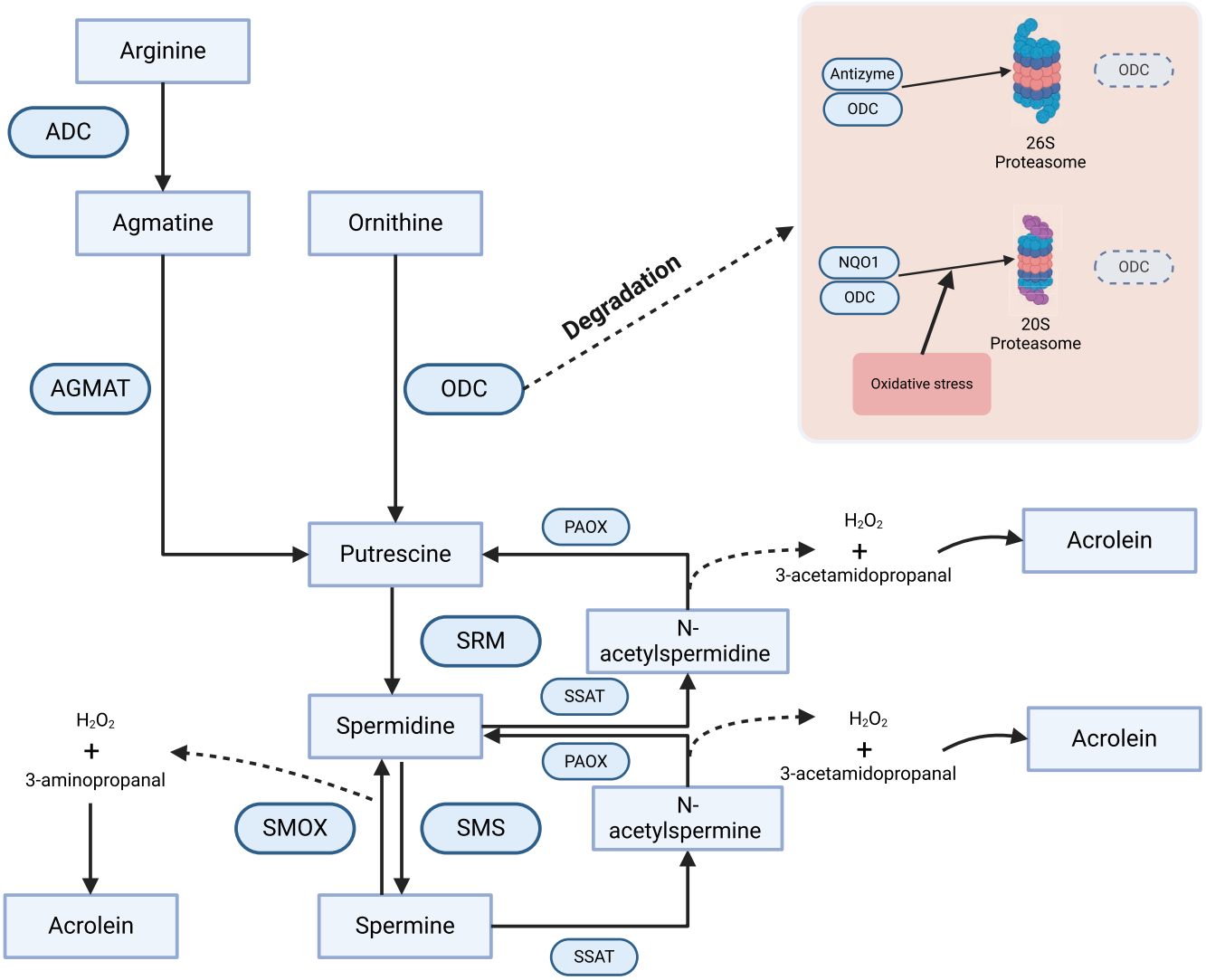


FIGURE 3 Overview of the polyamine synthesis pathway. The polyamine synthesis pathway starts with the formation of putrescine, either from ornithine by ODC or from arginine by ADC to first form agmatine and then by AGMAT (top-left). Putrescine can be converted to spermidine by SRM, after which SMS can act upon it to form spermine (bottom-middle). Spermine and spermidine can both be catabolized in a two-step process where it is first acetylated by SSAT and then oxidized by PAOX to form spermidine or putrescine. Spermine can also by directly oxidized by SMOX. These oxidation reactions generate H_2O_2 and 3-aminopropanal (by SMOX) or 3-acetamidopropanal (by PAOX). The 3-aminopropanal or 3-acetamidopropanal together with H_2O_2 form acrolein. ODC degradation is visualized within the red square (top-right). Degradation can occur either via ODC antizyme in a 26S proteasome dependent manner or via NQO1 in a 20S proteasome-dependent manner. The latter reaction is stimulated by oxidative stress. ADC, arginine decarboxylase; AGMAT, agmatinase; H_2O_2, hydrogen peroxide; NQO1, NAD(P)H quinone oxidoreductase 1; ODC, ornithine decarboxylase; PAOX, polyamine oxidase; SMOX, spermine oxidase; SMS, spermine synthase; SRM, spermidine synthase; SSAT, spermine/spermidine-N-acetyltransferase. Created in https://BioRender.com.

3-acetamidopropanal are formed (Alfarhan et al., 2020; Kashiwagi & Igarashi, 2023). Spermine can also be oxidized directly without the need for prior acetylation via spermine oxidase (*SMOX*), which generates hydrogen peroxide and 3-aminopropanal (Wang et al., 2014). Via a non-enzymatic reaction, both 3-acetamidopropanal and 3-aminopropanal can be converted to acrolein (Kashiwagi & Igarashi, 2023). Acrolein is a highly reactive and cytotoxic aldehyde that can induce protein adducts by conjugating with lysine (forming FDP-lysine). In addition, it can also induce oxidative stress by depleting cells of antioxidants and it is known to induce mitochondrial dysfunction (Alfarhan et al., 2020).

Activity of the polyamine synthesis pathway can be regulated through the degradation of ODC. Two reactions have been described that are involved in ODC degradation (see Figure 3, red square). The first and best-characterised reaction is the interaction between ODC and antizyme, which leads to 26S proteasomal degradation of ODC (Pegg, 2006). The second reaction requires the binding of NAD(P)H quinone oxidoreductase 1 (NQO1) to ODC, which enhances 20S proteasomal degradation. This NQO1-dependent form of ODC degradation is upregulated by oxidative stress.

Little is known regarding the effects of OAT deficiency on the polyamine synthesis pathway. One article reported increased urinary polyamines (predominantly putrescine) in GACR patients compared to healthy controls (Sulochana et al., 2000). In vitro data suggest a correlation between OAT dysfunction and RPE cell death that is dependent on the polyamine synthesis pathway (Kaneko, Okuda-Ashitaka, et al., 2007). The potential toxic effect of polyamines has also been demonstrated in healthy rats: injecting healthy Brown Norway rats intravitreally with

the polyamine spermidine resulted in oxidative stress-mediated RPE cell death. Injection of spermidine also induced increased oBRB permeability in these rats, as assessed by vitreous fluorophotometry (Ohashi et al., 2017). This permeability is in part explained by the disruption of the tight junctions between RPE cells, which are part of the oBRB. It is possible that these toxic effects induced by polyamines arise due to acrolein formation. Although currently no link has been identified between acrolein formation and GACR, acrolein-induced oxidative stress is a potential disease mechanism for diabetic retinopathy, another disorder affecting the retina (Alfarhan et al., 2020; Murata et al., 2020).

2.3 | The creatine synthesis pathway

An overview of the creatine pathway and how this relates to ornithine metabolism is presented in Figure 4. The synthesis of creatine starts in the kidney, where glycine and arginine are used to form guanidinoacetate and ornithine in a reaction catalysed by the enzyme AGAT. After transportation to the liver, guanidinoacetate is methylated by guanidinoacetate N-methyltransferase (GAMT), using a methyl group from S-adenosylmethionine (SAM), to form creatine and S-adenosylhomocysteine (SAH). After the formation of creatine in the liver, it can be transported via the blood to other tissues. Intracellular uptake of creatine is mediated by the creatine transporter SLC6A8. Creatine can be phosphorylated to form phosphocreatine via a reversible enzymatic reaction that requires adenosine triphosphate (ATP). Both creatine and phosphocreatine can be converted via non-enzymatic cyclisation to creatinine, which is excreted in urine (da Silva et al., 2009; Joncquel-Chevalier et al., 2015).

2.4 | The proline synthesis and degradation pathway

Proline is an amino acid involved in many different processes: it plays a role in protein synthesis, gene expression regulation, has antioxidant properties and it is a major component of collagen (Wu et al., 2011). Furthermore, proline can serve as a major source of energy when it is oxidized. An overview of the pathways involved with proline homeostasis is given in Figure 5.

OAT functions as a bidirectional mitochondrial enzyme, either consuming ornithine or synthesising it. OAT is widely expressed in virtually all human tissues, as

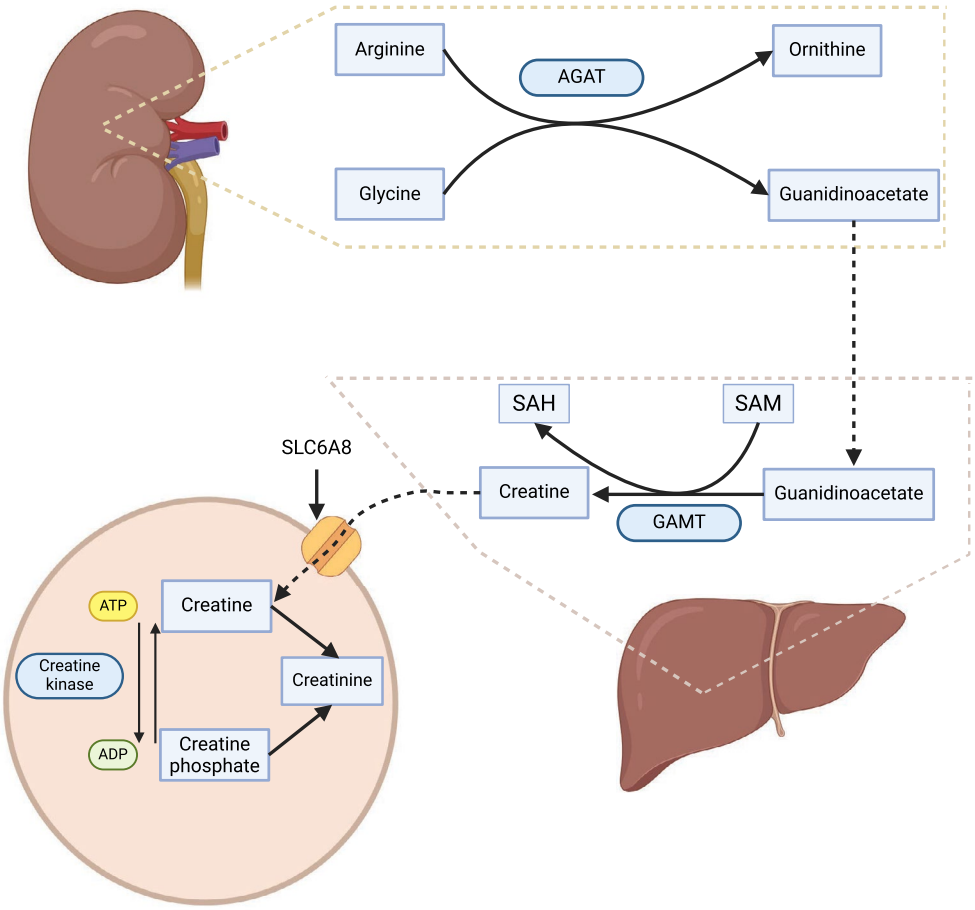


FIGURE 4 Overview of the creatine synthesis pathway in relation to ornithine metabolism in a wild type situation. On the top left, arginine and glycine are converted by AGAT to form ornithine and guanidinoacetate in the kidney. Guanidinoacetate is transported to the liver (bottom-right). In the liver GAMT uses guanidinoacetate and SAM to form creatine and SAH. Creatine is then transported to other tissues/cell types (bottom-left) where it is phosphorylated intracellularly by creatine kinase. Both creatine and creatine phosphate can non-enzymatically be converted to creatinine. The black arrows indicate (enzymatic) reactions, whereas the dashed black arrows indicate transport of a compound. ADP, adenosine diphosphate; AGAT, arginine:glycine amidinotransferase; ATP, adenosine triphosphate; SAH, S-adenosylhomocysteine; SAM, S-adenosylmethionine. Created in https://BioRender.com.

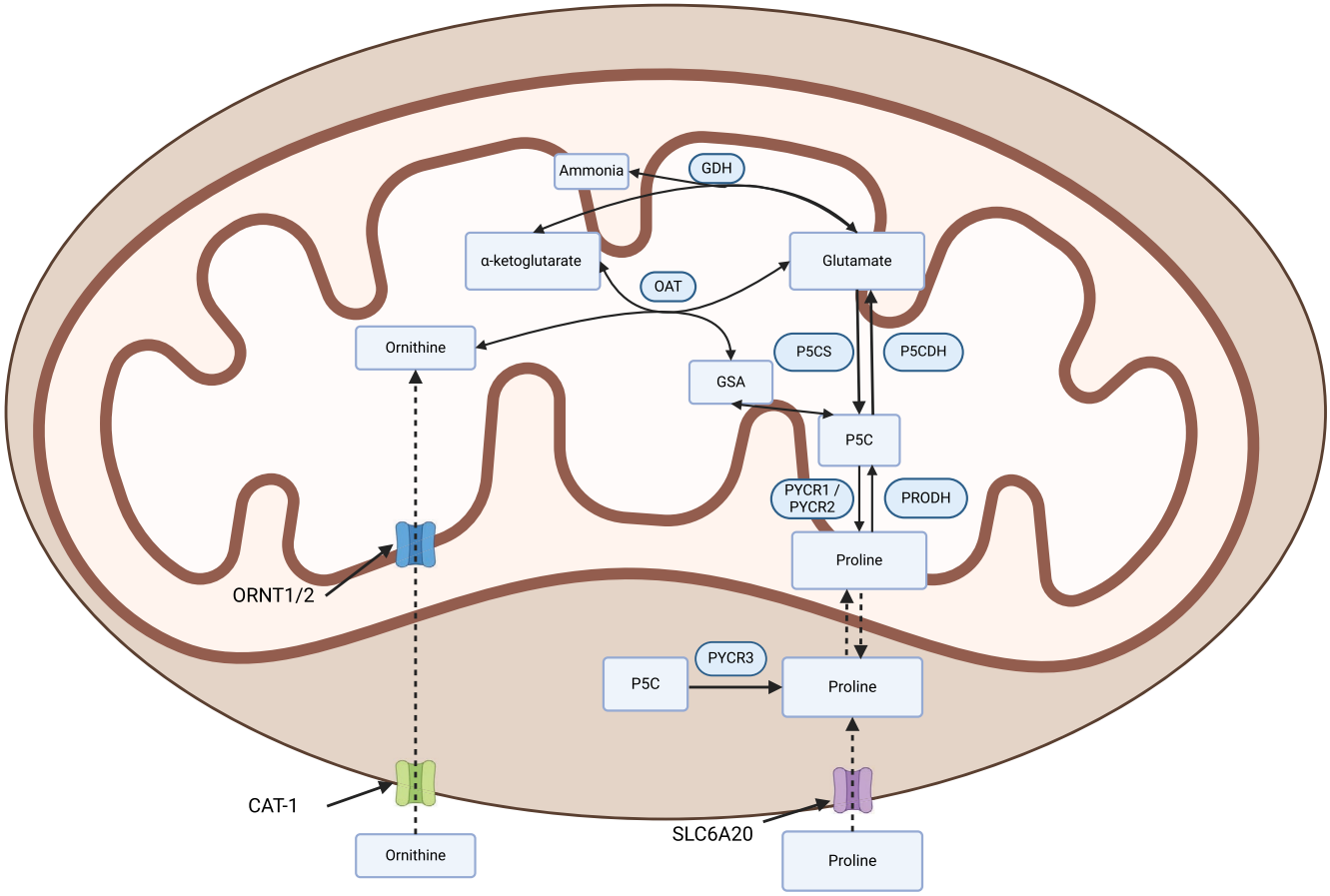


FIGURE 5 Overview of the proline synthesis and degradation pathway. Within the mitochondria, ornithine and α-ketoglutarate are converted to glutamate and GSA in a reaction catalysed by OAT (left). GSA can be converted to P5C. P5C can either be used to form proline by PYCR1/PYCR2 or to form glutamate by P5CDH. Glutamate can also be formed by a deamination reaction catalysed by GLS, which also forms ammonia (top-right). From glutamate, it is possible to generate P5C (via P5CS) or α-ketoglutarate and ammonia (via GDH). Proline can enter the cell via the SLC6A20 transporter and can also be formed in the cytosol from P5C by PYCR3 (bottom). CAT-1, cationic amino acid transporter-1; GDH, glutamate dehydrogenase; GLS, glutaminase; GSA, glutamate-5-semialdehyde; OAT, ornithine aminotransferase; P5C, pyrroline-5 carboxylate; P5CDH, pyrroline-5-carboxylate dehydrogenase; P5CS, pyrroline-5-carboxylate synthetase; PRODH, proline dehydrogenase; PYCR1, pyrroline-5-carboxylate reductase 1; PYCR2, pyrroline-5-carboxylate reductase 2; PYCR3, pyrroline-5-carboxylate reductase 3. Created in https://BioRender.com.

concluded from studies at the mRNA and protein level. Tissues with the highest abundance of OAT protein are the kidney, liver, intestine and retina. It is important to note that OAT expression is not ubiquitous within the liver, where it can be detected mostly in the pericentral hepatocytes and not in periportal hepatocytes, in which the urea cycle takes place (Ginguay et al., 2017). Evidence from animal studies suggests that, unlike many other tissues, intestinal OAT synthesises ornithine from P5C (Henslee & Jones, 1982; Ross et al., 1978). Importantly, several post-translational modifications of OAT have been identified, which include PLP cofactor fixation, *N*-succinylation and *N*-acetylation. PLP fixation is an essential OAT modification to ensure proper transaminase function; however, how *N*-succinylation and *N*-acetylation affect OAT is still unknown (Ginguay et al., 2017).

In most tissues, OAT catalyses the breakdown of ornithine and α-ketoglutarate into glutamate and glutamate-5-semialdehyde (GSA). GSA can be spontaneously converted to P5C, from which proline can be generated by pyrroline-5-carboxylate reductase (PYCR). Three PYCR isoforms exist; PYCR1 and PYCR2 are both localized to the mitochondria, whereas PYCR3 is in the cytosol (De Ingeniis et al., 2012). Proline can also be synthesized from glutamate when glutamate is first converted to P5C by P5CS. Due to alternative splicing, a long and short isoform of P5CS exist. The short isoform is only expressed in the gut and is inhibited by ornithine, whereas the long isoform is expressed in multiple tissues but is not inhibited by ornithine (Perez-Arellano et al., 2010).

The first step in proline degradation is catalysed by proline dehydrogenase (PRODH) to form P5C, then pyrroline-5-carboxylate dehydrogenase (P5CDH) generates glutamate, which finally is converted to α-ketoglutarate and ammonia via either glutamate dehydrogenase (GDH) or GOT2 (Patriarca et al., 2021). α-Ketoglutarate can then enter the tricarboxylic acid cycle to generate energy in the form of ATP, NADH and FADH or can be used for the synthesis of citrate by means of a process called reductive carboxylation. Under hypoxic conditions, it is also possible that α-ketoglutarate derived from glutamate can be converted to isocitrate via IDH2 (Wise et al., 2011). Glutamate can also be used to synthesise glutamine via glutamine synthetase (GS), although this enzyme is only expressed in astrocytes and liver pericentral hepatocytes (Schousboe et al., 2014). Different inherited metabolic disorders, in

addition to GACR, with defects in the proline synthesis and degradation pathway have been identified (Garanto et al., 2022). As an example, cutis laxa patients caused by P5CS deficiency also display retinal abnormalities and early cataract formation (Wolthuis et al., 2014). It is currently unknown to what extent the proline synthesis and degradation pathway is affected in GACR because so far, no other metabolite in this pathway other than ornithine appears to be altered.

3 | UNRAVELLING THE RETINAL PATHOLOGY IN GACR

The retina contains several neural layers (consisting of photoreceptors, horizontal cells, bipolar cells, amacrine cells and ganglion cells), the RPE and the vascular choroid lining the back of the eye. The neural retina is dependent on the RPE to function properly (Figure 6).

In GACR patients, the highest level of atrophy occurs in the choroid, RPE and the photoreceptors. Anecdotal evidence from one histopathological study of a GACR donor eye suggests that subclinical mitochondrial abnormalities are also present in several different ocular cell types, including ciliary body epithelia, corneal endothelium, smooth muscle of the iris and ciliary body (Wilson et al., 1991). In GACR, the primary site of damage is thought to be the RPE (Wang et al., 1996). Therefore, to understand the pathogenic mechanism at play in GACR, it is essential to have an in-depth understanding of the many functions of the RPE and its interaction with adjacent tissues, as described in this section. First, we describe the structure and function of the healthy RPE, also in relation to the adjacent choroid and photoreceptors. Next, we explore specific GACR-relevant features of RPE cells based on gene expression and function. Subsequently, we focus on GACR and ornithine metabolism-relevant features of the RPE cell. Finally, we will elaborate on well-established biochemical RPE pathological mechanisms.

3.1 | Healthy RPE cell structure and function; interaction with choroid and photoreceptors

An overview of the RPE cell structure, function and its interaction with adjacent cell types is presented in Figure 6. One of the main structural aspects of the RPE is that it forms the oBRB; the functional properties of the RPE are the transportation of nutrients, maintenance

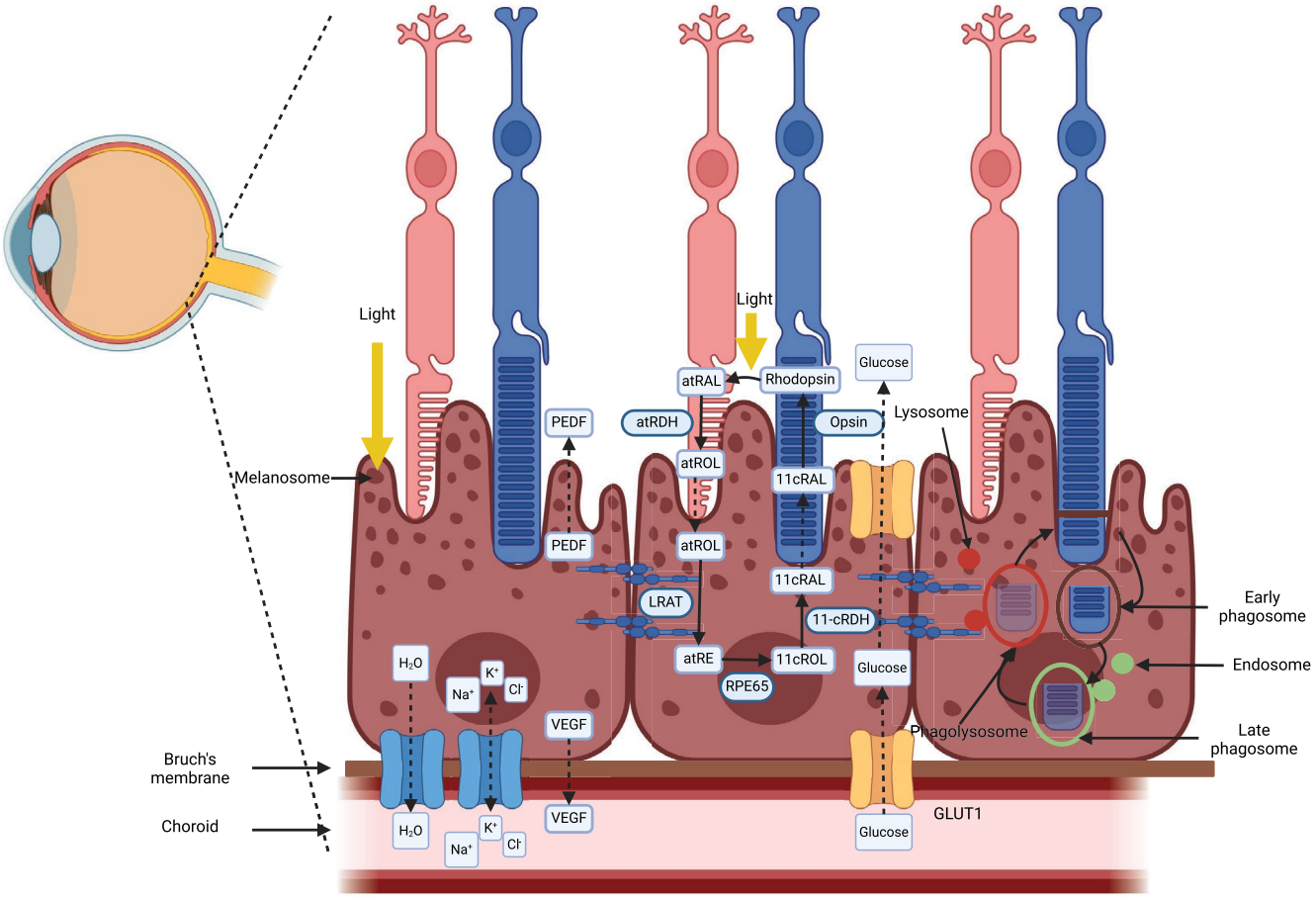


FIGURE 6 Schematic representation of different RPE cell functions. Melanosomes, present within the RPE cell, are able to absorb UV light in order to protect from photooxidation (top-left). Below the choroid is given, which is in contact with the Bruch's membrane and RPE cells. Transporters situated on the basal side of the RPE cell allow transport of nutrients and the secretion of growth factors between the RPE and choroid. On the apical side the RPE cells are in contact with photoreceptors. In the middle RPE cell and photoreceptors, the visual cycle is visualized. On the right side the process of photoreceptor outer segment phagocytosis is given. First the early phagosome is formed, which after fusion with endosomes (green) and subsequent fusion with lysosomes (red) can be broken down in nutrients and recycled back to the photoreceptor. Created in https://BioRender.com.

of the visual cycle, protection against photooxidation, (growth)factor secretion and the phagocytosis of photoreceptor outer segments (POS).

The RPE consists of a monolayer of cuboidal neuroepithelial cells present between the choroid/Bruch's membrane and the neuroretina. The RPE is a highly polarized cell type of which the apical side can interact with the POS, whereas the basal side is connected to the Bruch's membrane and choroid. RPE cells are linked together by tight junctions, thereby forming the oBRB. Formation of the oBRB is essential to create a physical barrier between the retina and the vascular system. This ensures an optimal environment for proper retinal functioning (Naylor et al., 2019).

A healthy RPE cell layer is crucial for the facilitation of the transport of nutrients from the blood to the retina, and for the removal of water and metabolic waste out of the retina. The RPE layer is also involved in the maintenance of the visual cycle by regenerating 11-*cis*-retinal from all-*trans*-retinol exported by the photoreceptors. It is also possible for 11-*cis*-retinal to be supplied to the RPE cell via the choroid. Subsequently, 11-*cis*-retinal is transported to the photoreceptors where it is combined with opsin to form the light sensitive protein rhodopsin (see Figure 6). Exposure of rhodopsin to light results in photobleaching and a repeat of the visual cycle (Strauss, 2005).

Another functional role of the RPE is to protect against photooxidation and counteract the high levels of oxidative stress in the retina due to constant exposure to light and daily digestion of POS. Even though most of the harmful UV light is being filtered out by the lens, the RPE is also able to absorb some UV light by the melanin stored within melanosomes in the RPE (D'Alba & Shawkey, 2019; Hejtmancik et al., 2015).

Another main function of the RPE is the secretion of many (growth)factors, which are important for the survival of photoreceptors and conserve the integrity of the retina and choroid. Vascular endothelial growth factor (VEGF) is primarily secreted basally, whereas pigment epithelium derived factor (PEDF) is secreted apically (Chen et al., 2022). Secretion of VEGF by the RPE is essential for the formation of choriocapillaris. Dysregulation of VEGF expression leads to choroidal neovascularization, which is observed in age-related macular degeneration (AMD) (Yeo et al., 2019). PEDF expression by the RPE has been shown to be protective for photoreceptors and retinal neurons against oxidative stress-induced damage (Wang et al., 2019).

Finally, RPE ensures proper photoreceptor functionality by the process of phagocytosis, in which damaged POS are renewed. To facilitate this process, the RPE contains huge amounts of endosomes and lysosomes (Lakkaraju et al., 2020). The phagocytotic process starts with the shedding of the POS. The RPE phagocytoses the POS to form a phagosome, which matures by first fusing with endosomes and then lysosomes to become a phagolysosome. After breaking down most of the components of the POS, the RPE returns nutrients such as docosahexaenoic acid and membrane-associated elements back to the photoreceptor (Kwon & Freeman, 2020; Mazzoni et al., 2014; Strauss, 2005).

3.2 | Expression of RPE signature genes and their relationship with ornithine metabolism

To gain insight into the genetic metabolic profile of the RPE, we assessed the signature gene expression profile of the RPE and its potential role in ornithine metabolism and GACR. Transcriptomic analysis on human fetal and adult RPE revealed a set of 154 signature RPE genes. Many of these genes were involved in transporter activity; vitamin A metabolism (*RPE65*, *BEST1*, *TIMP3*) and oxidoreductases (*ALDH1A3, PCYOX1, FADS1*, *CDO1*) (Strunnikova et al., 2010). Using laser microdissection, Booij et al. (2010) were able to specifically isolate RPE cells from donor eyes, without contamination of either photoreceptors or choroidal cells. By comparing RPE gene expression to the photoreceptor and choroidal samples, a set of 114 RPE-specific genes was identified, which were enriched for retinol metabolism, gamma-aminobutyric acid (GABA) signalling and the retinoic acid receptor activation pathways (Booij et al., 2010). Finally, by comparing RPE samples from mice and humans, 22 genes were found to be highly expressed in both samples. In addition to genes related to the visual cycle, several transporter genes were represented (*SLC6A13, SLC6A20, SLC7A10, SLC16A8, SLC39A12*) (Bennis et al., 2015). The fact that these five transporter genes are highly expressed in both human and mouse RPE merits further exploration. The SLC6 family of transporters is known to be involved in the transport of neurotransmitters, metabolites involved in energy production and amino acids. *SLC6A13* is a chloride and sodium-dependent transporter that can transport GABA, taurine, hypotaurine and beta-alanine across the membrane (Rudnick et al., 2014). The uptake of proline into cells is primarily carried out by the plasma membrane transporter SLC6A20. In addition, the SLC6A20 transporter also participates in glycine transport and modulation of the NMDA receptor (a glutamate receptor) (Bae et al., 2021). SLC16A8 is a lactate transporter that has been shown to be expressed in the basolateral membrane of the RPE specifically. Furthermore, variations in the encoding gene have been associated with retinal degenerative disorders such as AMD (Fritsche et al., 2016). *SLC7A10* encodes for an antiporter responsible for the transport of glycine, serine and the neutral amino acids alanine and cysteine. Studies have shown that this protein plays a key role in the glycinergic inhibitory system (Ehmsen et al., 2016; Safory et al., 2015). Mutations in the encoding gene have also been associated with retinal degenerative disorders such as AMD. Finally, *SLC39A12* encodes a zinc transporter. Zinc has an integral role in the metabolism of many enzymes, lipids, nucleic acids and carbohydrates, in addition to its role in modulating gene expression (Scarr et al., 2016). The supplementation of zinc has been a topic of interest for many retinal degenerative disorders since zinc can reduce oxidative stress and slow down AMD progression (Blasiak et al., 2020).

3.3 | Amino acid metabolism and metabolic interconnection between RPE and photoreceptors

The RPE is a highly specialised cell type, with tightly regulated metabolic pathways. Changes in RPE metabolism have been correlated with ophthalmic disease progression. By performing metabolomics and lipidomic analyses in AMD-affected RPE compared to healthy RPE, several metabolites were found to be differentially regulated. Among these metabolites are glutathione, lipids involved in autophagy and L-glutamic acid (Zhang et al., 2020). Dysfunction of RPE metabolism, induced by mitochondrial oxidative stress, has also been linked to photoreceptor deterioration secondary to RPE dysfunction (Brown et al., 2019).

In this section, we will discuss the metabolism of ornithine within the RPE, the metabolism of proline within the RPE and the energy production pathway interconnection between the RPE and photoreceptors.

3.3.1 | Ornithine metabolism within the RPE

OAT expression is relatively high in RPE cells, indicating the importance of ornithine metabolism in this cell type. Several ornithine transporters have been identified. Specifically, the cationic amino acid transporter (CAT-1) encoded by *SLC7A1* was shown to be involved in ornithine-induced toxicity in OAT-deficient RPE cells (Kaneko, Ando, et al., 2007). Silencing of this ornithine transporter in vitro in RPE cells is sufficient to limit ornithine-induced toxicity in a GACR model (see Section 5).

One major aspect of ornithine metabolism is different in the RPE when compared to the previously described pathways. Protein expression of OTC, necessary for the conversion of ornithine to citrulline, is primarily located in the liver and intestine (Couchet et al., 2021). As RPE cells lack expression of the OTC enzyme, they are unable to initiate the urea cycle and thus rely heavily on either the polyamine synthesis pathway or proline and/or glutamate synthesis pathways to degrade excess ornithine (Koshiyama et al., 2000).

3.3.2 | Proline metabolism within the RPE

The RPE depends heavily on proline as a source of energy compared to other tissues (Chao et al., 2017). In the healthy RPE, proline catabolism is not only essential for the generation of energy via the OXPHOS system thereby preserving glucose for preferential usage by photoreceptor cells but is also important for the generation of glutathione, glutamine, citrate, other amino acids, and numerous mitochondrial intermediates which are subsequently transported to photoreceptor cells. Upon maturation, *in vitro* RPE cells consume more proline (Yam et al., 2019). Combined human and mouse retinal cell studies in vitro suggested that proline catabolism by the RPE results in the production of glutamine, glutamate and aspartate, which are subsequently taken up by the neural retina. Overabundance of PRODH in the RPE compared to the retina further supports the notion that the RPE cells prefer to catabolize proline (Zhu et al., 2023).

In situations where there is reduced availability of proline, a lack of intracellular glutathione in the RPE might arise. Glutathione has strong anti-oxidant and immune system activating properties (Sreekumar et al., 2021). Indeed, reduced intracellular glutathione availability has been linked to the induction of ferroptosis, autophagy and stress-induced premature senescence in RPE cells (Sun et al., 2018). Proline supplementation enhances expression of the reductive carboxylation pathway and therefore the generation of glutathione is stimulated to protect the RPE from oxidative stress-induced damage. The importance of mitochondrial metabolism for the RPE is further emphasized by the fact that oxidative stress is the main driver of other retinal diseases, such as AMD and diabetic retinopathy (Ferrington et al., 2020).

3.3.3 | Energy production pathways and the interconnection between RPE and photoreceptors

The RPE mainly relies on oxidative phosphorylation for energy production, whereas the photoreceptors use aerobic glycolysis (Brown et al., 2019). Glucose can be transported through the RPE cell via the GLUT1 receptor, which is present both basally and apically, to the photoreceptors. The photoreceptors consume glucose via anaerobic glycolysis, thereby generating lactate. This lactate is transported back to the RPE via monocarboxylate transporter 1 (MCT1, encoded by *SLC16A1*) where it can be converted to pyruvate (Hurley, 2021; Lakkaraju et al., 2020). In addition to lactate, the RPE also prefers to use proline oxidation as a source of energy. The RPE consumes high amounts of proline while the neural retina utilises the proline-derived intermediates and glucose.

3.4 | Oxidative stress and the RPE

The retina, and especially the RPE, is one of the most oxygen-consuming tissues of the human body (Du et al., 2021). The RPE not only absorbs photons of incoming excess light, but also needs to daily phagocytose POS. Consequently, these RPE activities generate huge amounts of local oxidative stress and thus ROS. These ROS can react with DNA, proteins, lipids and membranes and induce damage (Pizzino et al., 2017). Oxidative stress-induced damage is a major driver of RPE dysfunction in retinal disorders, such as AMD and proliferative vitreoretinopathy (Du et al., 2021).

This overload of local oxidative stress has been linked to the dependence of the RPE on reductive carboxylation (Du et al., 2021). Reductive carboxylation is the process by which α-ketoglutarate is converted to citrate and NADP+ via IDH2. Citrate can be transported from the mitochondria to the cytosol where either citrate lyase uses it to generate acetyl coenzyme-A for fatty acid synthesis, or IDH1 uses it as a substrate to

form α-ketoglutarate and NADPH. Cytosolic NADPH is essential for protection against oxidative stress, as it can reduce glutathione to perform its antioxidant function (Gaucher et al., 2018). By supplementing RPE cells exposed to oxidative stress with either α-ketoglutarate or an NAD^+ precursor, reductive carboxylation activity could be sustained and therefore the cell viability of the cells could be protected (Du et al., 2016).

High levels of ROS are generated in the mitochondria, and predominantly by the mitochondrial electron transport chain, which is coupled to oxidative phosphorylation (Zhao et al., 2019). RPE cells use the oxidative phosphorylation pathway to use lactate as a source for generating ATP. Lactate also inhibits glycolysis within the RPE, ensuring sufficient transport of glucose to the photoreceptors (Kanow et al., 2017). Among others, the impact of mitochondrially generated oxidative stress on the RPE was illustrated by studies of a manganese superoxide dismutase Sod2−/− mouse model. SOD2 is one of the key ROS detoxifying enzymes within the mitochondria. Knocking out Sod2 in this mouse model resulted in RPE dysfunction, which subsequently affected photoreceptor mitochondria due to increasing RPE glycolytic metabolism (Brown et al., 2019).

4 | WHY IS THE RETINA PRIMARILY AFFECTED in GACR?

Despite increasing insights into RPE (patho)physiology, the reason why the retina is primarily affected in GACR remains a mystery. *OAT* is expressed in virtually every human tissue, and all cells need OAT activity for proper mitochondrial function, energy production and amino acid homeostasis. There may be several (combinations) of reasons why the retina, and most notably the RPE, is primarily affected in GACR. These include molecular, cellular and/or systemic differences between RPE and other cells/tissues in the body. Differences in *OAT* splice variants, post-transcriptional modifications or protein expression could have implications for OAT functionality in specific tissues. Its (bi-directional) activity may depend on cell type-specific gene expression, protein activity and metabolic substrates or products of several other ornithine-related pathways; these pathways should also be considered.

In this section, we will discuss several hypotheses to explain the GACR phenotype.

4.1 | Does the unique dependency of the RPE on proline play a role in GACR?

One unique aspect of the RPE (compared to other cell types) is its reliance on proline catabolism for energy production. Proline can be obtained from different sources including the diet, endogenous synthesis from either glutamate or ornithine (see Figure 5) or through the breakdown of collagen. Proline may enter RPE via the SLC6A20 transporter. It is possible that the high levels of ornithine in the case of GACR somehow interfere with the SLC6A20-mediated uptake of proline into RPE cells; however, this should be further investigated.

Indeed, the RPE uses proline both for reductive carboxylation and the generation of intermediates for mitochondrial oxidative phosphorylation (Yam et al., 2019). The importance of proline for the RPE is highlighted by RPE-specific expression of the proline transporter SLC6A20, and the fact that the RPE cells consume more proline as they mature. Adhering to a proline-rich diet has even been shown to suppress induction of retinal degeneration in a mouse model injected with sodium iodate to selectively induce oxidative stress-related RPE cell death (Yam et al., 2019). Since one of the proline-producing pathways is directly disturbed in GACR (see Section 2.4), a lack of proline availability in the RPE and retina could be considered a likely risk factor for retinal damage observed in GACR. However, amino acid analysis of GACR patient plasma has not identified decreased levels of proline availability (Zekusic et al., 2018). Consequently, it is currently not clear whether proline supplementation would be beneficial for GACR patients.

4.2 | Could activation of the polyamine synthesis pathway explain the pathological mechanism of GACR?

As described in Section 3.3.1, ornithine-induced toxicity in human RPE cells in vitro could be inhibited by silencing the ornithine transporter CAT-1 located in the plasma membrane. This suggests that ornithine needs to be imported into the cell before toxicity arises. In the aforementioned in vitro model, exposing RPE cells to ornithine and inhibiting OAT activity resulted in the upregulation of ODC. This coincided with increased synthesis of the polyamines spermidine and spermine, which, in turn, both induce upregulation of CAT-1 and ODC gene expression (Kaneko, Okuda-Ashitaka, et al., 2007). These results were expanded upon in a rat model that received an intravitreal administration of spermidine. This administration was sufficient to induce RPE cell death, which is attributed to the generation of hydrogen peroxide and acrolein (Ohashi et al., 2017). It can be hypothesised that in GACR RPE cells, where there is no expression of either OAT or OTC, polyamine synthesis is likely further enhanced, resulting in toxic levels. In this case, oxidative stress would be generated by the formation of hydrogen peroxide and acrolein, specifically in the RPE.

4.3 | Once damaged, the barrier function of the RPE in GACR cannot be restored

A key cellular difference between the tissues with the highest level of OAT expression, being liver, intestine, kidney and retina (predominantly RPE) is their ability to proliferate. Even though cells in the liver are quiescent in a healthy state, when damaged, the liver can restore substantial amounts of damaged tissue (Bangru & Kalsotra, 2020). Similarly, the intestine and kidney are well known for their cellular proliferation and regenerative properties (Guevara-Garcia et al., 2023; Wu et al., 2021). In contrast, when injury occurs to the RPE,

the prevailing hypothesis is that RPE cells in vivo elongate to 'close the gap' to maintain the integrity of the barrier and interaction with neighbouring RPE cells. Once RPE cells 'meet' they become quiescent due to contact inhibition. The required enlargement of these RPE cells however comes at the cost of (temporary) reduced expression of mature RPE cell markers and decreased expression of tight junction proteins (George et al., 2021). It can be hypothesized that chronic ornithine exposure, without the capability of cell renewal, is a reason why the RPE cells are more affected than liver, kidney or intestinal cells. Chronic ornithine-induced toxicity may affect proper tight junction function, thereby further sensitising the RPE cell to ornithine exposure. In a rat model of acute pancreatitis, administration of l-ornithine directly affected blood–brain barrier integrity. This was also found in an in vitro rat blood–brain barrier model, where L-ornithine administration not only induced oxidative stress, but also led to increased permeability of the barrier (Walter et al., 2022).

4.4 | Is there cumulative damage of the photoreceptor-RPE-choroid complex in GACR?

The photoreceptor–RPE–choroid complex in the retina forms one functional unit. Thus, dysfunction of one cell type directly influences the function and viability of the other cell types. For example, damage to the RPE itself may not only lead to RPE dysfunction, but may also invoke disruption of the oBRB. This could subsequently not only lead to damaging neovascularization, but also to increased retinal influx of hazardous compounds, among which ornithine, to enter the neural retina where they induce additional damage. Obviously, damage to the nervous tissue immediately leads to irreversible loss of function, as the neural retina, including the photoreceptors, consists of fully differentiated non-replicating cells.

As described in Section 3.3.1, dysregulation of metabolic pathways in either the RPE or photoreceptors has a severe impact on the other cell type. As the oBRB functions as a strict gateway, making it difficult for nutrients and proteins to enter the eye, it is of utmost importance that metabolic functioning within the eye is properly regulated. This could explain why patients with inherited metabolic disorders also often experience ophthalmological symptoms (Ferreira et al., 2023).

5 | GYRATE ATROPHY RESEARCH, WHERE TO GO FROM HERE?

Although current treatment modalities might slow down disease progression, affected individuals will eventually develop severe visual impairment (Kaiser-Kupfer et al., 2002, 2004). Therefore, optimisation of treatment is warranted. However, as there are indications that early treatment leads to better outcomes, it is important to consider improving diagnosis. Most patients are diagnosed in the second decade of life and often already experience chorioretinal atrophy at the moment of diagnosis (Balfoort, Van Den Broeck, et al., 2024). Including GACR in newborn screening would allow for early diagnosis and thus early treatment. Due to reversed OAT flux in the neonatal phase, ornithine is not a sufficient biomarker for newborn screening, but a potential candidate could be the proline/citrulline ratio (de Sain-van der Velden et al., 2012).

As GACR is such a rare disease, clinical reporting has not been consistent, with published case reports from different institutions all reporting different parameters to determine GACR disease progression. Because of this, and the heterogeneity in disease progression, much remains to be learned about the effect of OAT dysfunction on retinal degeneration. To solve this issue, the Gyrate Atrophy Ocular and Systemic (GYROS) Study was started late 2023 (NCT05312736). The GYROS study is a natural history study that over a period of 4 years aims to characterise retinal degeneration, ornithine levels (while following the diet), patient variability, metabolic-structure–function relationships and identify rapid progressors. Forty-five patients from across the world are enrolled within this study. In a field where most available patient data consists of single patient case reports, a study of this size and duration will provide valuable insights into the disease progression of GACR. Furthermore, a cohort study of 27 GACR patients providing insights on GACR treatment effects, biochemistry, genotype and phenotype has been completed recently. This study consists of patients from different ethnicities and yields even more evidence regarding the effect of the protein-restricted diet on retaining visual fields (Balfoort et al., 2025).

Multiple pre-clinical GACR models have been constructed and studied. These can be largely divided into cell lines and animal models. More specifically, these include RPE or other cell types in which OAT activity is inhibited by chemical or molecular genetic tools, or genetically modified animal models. We describe these models and their usefulness for (experimental) GACR therapeutics below.

Cultured primary human RPE cell lines have been used to study GACR in vitro. Ito and coworkers applied 5-fluoromethylornithine (5-FMO), a specific inhibitor of OAT, to mimic GACR (Ueda et al., 1998). The authors concluded that ornithine or 5-FMO alone did not damage RPE cells, but combined they did. Furthermore, they demonstrated that cytotoxicity induced by the combination of 5-FMO and ornithine was prevented by the addition of a series of amino acids equipped with a nonpolar side chain (leucine, isoleucine, valine, methionine, alanine, proline, phenylalanine and glycine) or an uncharged polar side chain (glutamine, asparagine, serine and threonine) to the RPE cell culture medium (Nakauchi et al., 2003). In contrast, the two charged amino acids: glutamate and aspartate did not inhibit cytotoxicity.

Interestingly, the same researchers were also able to reduce ornithine-induced toxicity of RPE cells by siRNA-mediated inhibition of CAT-1 (Kaneko, Ando, et al., 2007). CAT-1 expression was targeted because its expression in RPE was upregulated by combined OAT inhibition and ornithine supplementation.

Next to inhibiting CAT-1 expression directly, a reduction could also be achieved biochemically by inhibiting ODC, using alpha-difluoromethylornithine (DFMO). This inhibits the potentially toxic polyamine synthesis pathway (Kaneko, Okuda-Ashitaka, et al., 2007) (see Section 2.2). Reducing CAT-1 expression by inhibiting ODC was sufficient to prevent ornithine and 5-FMO induced toxicity in RPE cells. 5-FMO has also been used in mice and chickens. Long-term treatment with this compound resulted in elevated ornithine levels in the eye of these animals, but no toxic effects were observed (Daune-Anglard et al., 1993).

Several OAT-deficient cell lines have been generated and used. In the ARPE-19 cell line, *OAT* was knocked down using siRNA. Even though OAT activity was largely abolished, the addition of ornithine and/or arginine did not induce cytotoxicity within the timeframe of the experiments (Montioli, Bellezza, et al., 2021). This group also generated an OAT-deficient HEK293 cell line in which specific human *OAT* variants (Q90E, R154L, G237D, R271K, E318K and C394Y) were subsequently expressed by lentiviral transduction to study structural folding defects and functional deficiencies of the OAT enzyme (Montioli, Sgaravizzi, et al., 2021). Finally, an induced pluripotent stem cell (iPSC) line was derived from a GACR patient. This iPSC line harbours the A226V variant, which has been described as vitamin B6-responsive (Michaud et al., 1995). This iPSC line was then differentiated into RPE cells, and when pyridoxine was added to these RPE cells, OAT enzymatic activity was greatly enhanced (Meyer et al., 2011).

In addition to the aforementioned in vitro and in vivo models, a genetic OAT-deficient mouse model has been generated. This mouse model developed hypo-ornithinaemia, resulting in death within 48 h after birth. This hypo-ornithinaemia is due to the fact that in the neonatal phase, the OAT enzyme is used for ornithine generation instead of consumption, similar to humans. The OAT-deficient mouse is thus unable to generate sufficient amounts of ornithine. The finding that the lethality caused by hypo-ornithinaemia could be prevented by intraperitoneal arginine injection allowed the mice to survive the neonatal period (Wang et al., 1995). When these surviving mice were placed on an arginine-restricted diet and evaluated for a period of over 12 months by retinal histological and ultrastructural studies, ERG and by measuring plasma amino acids, plasma ornithine levels decreased and retinal degeneration could be completely prevented (Wang et al., 2000). These results still form the basis for current GACR therapy.

More recently, the effect of liver-directed gene therapy was studied using the *Oat*^{rhg} mouse model which has a homozygous missense variant in Oat (p.Gly353Ala). By restoring OAT activity specifically in the hepatocytes via delivery with adeno-associated virus serotype 8 (AAV8), it was possible to reduce the ornithine concentration both in the blood and eye cups. Furthermore, ERG responses improved and there was partial restoration of the retinal structure which was still present 1 year after injection in the AAV8-treated mice compared to the untreated ones (Boffa et al., 2023). This study invoked the discussion on whether targeting the eye or liver would be optimal for treatment of GACR (Bergen et al., 2024; Boffa & Brunetti-Pierri, 2024).

Recent advances in the field of ophthalmic research allow us to use new scientific methods to identify the exact pathological mechanism in GACR and ideally indicate how to counteract this. Generation of retinal organoids and stem cell-derived RPE from patients and exposing them to ornithine provides the opportunity to completely characterize the course of the disease on a cellular, molecular and genetic level (Wagstaff et al., 2021). It is of utmost importance to determine the dysregulation of all pathways described in this review in these in vitro models, and combine new insights with previous research results. Furthermore, efforts should be made to compare this retinal organoid model with liver, intestine and kidney organoids, as these tissues also express high levels of OAT. Organoid models for all these tissues exist and improvements are made towards generating organoids that are more representative of the in vivo situation at a very high rate (Harrison et al., 2021; Khoshdel-Rad et al., 2022; Taelman et al., 2022). Generation of these models will allow us to test a plethora of potential protective supplements in a safe setting before applying them in animal models or prescribing them to GACR patients.

It is important to note that all described models have their own strengths and weaknesses. The in vitro models (OAT deficient either due to 5-FMO or genetic deficiency) allow a range of experiments in a highly controlled environment where the direct impact of OAT deficiency on the cell type can be studied. These models, however, lack the complexity of the interaction of different cell types and organ systems, which is a key component in GACR pathology. This problem can, to some extent, be resolved by using retinal organoids or stem cell-derived RPE, but this increased complexity does result in reduced reproducibility and increased costs and time duration of the experiments. Moreover, the animal models (especially the mouse model) allow for long-term experiments that can investigate the effect of OAT deficiency and therapeutic intervention in a model that more closely resembles the patient in vivo situation. Despite the fact that the mouse model has inherent limitations, for example, significant differences between the expression of metabolic pathways and the structure of the eye between mice and humans, such models remain critical to understand inter-organ communication and metabolic fluxes in a metabolic disorder such as GACR.

Several revolutionary treatment modalities have been developed that could be of interest for future GACR treatment. As we described in our recent publication, we believe that specific targeting of the eye with gene therapy, possibly in combination with other strategies to lower systemic ornithine, would be the most valuable therapy option to pursue (Bergen et al., 2024). The application of gene therapy for the treatment of monogenetic retinal disorders has already been applied very successfully (Russell et al., 2017). By only lowering ornithine concentrations, retinal cell death cannot be prevented. Restoring ornithine metabolism within the eye and the RPE seems essential when you critically examine how

important these pathways are for proper RPE cell functioning. Another method is optogenetics, which is considered promising in the field of neuroscience and retinal degeneration. Optogenetics is a mutation-independent form of therapy that aims to convert cells in the inner retinal layer (e.g. bipolar cell) to become artificial photoreceptors. This is particularly useful in advanced retinal degenerative disorders where the photoreceptors have died and thus cannot be targeted by other treatment options (Sahel et al., 2021; Sakai et al., 2022). Finally, cell replacement therapy could also benefit patients with more advanced GACR pathology. Different clinical trials are currently ongoing using RPE cell replacement (Sanie-Jahromi & Nowroozzadeh, 2022). This form of therapy shows promise in the preclinical stages, but further research and optimisations are still needed before it can be a viable treatment option for GACR patients (Wu et al., 2024).

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